

M. M. Institute of Computer Technology & Business Management (MCA), M.M. (Deemed To Be University), Mullana			Course Coverage Plan (Practical)
Session: 2025-26 (Even)		Class/Semester: BCA – 6 th Semester (G1 & G2 & G3)	
Subject Code: BCA-609		Subject Name: Data Visualization Lab	
Lecture: 2		Tutorial: 0	
Max. Marks Theory: 60		Max. Marks Sessional: 40	Credit 1.0
Lecture No.	G1&G2	G3	Topic Discussed
1	08-01-26	09-01-26	Create a simple bar chart using student marks or sales data and explain what it shows.
2	08-01-26	09-01-26	Draw a line chart to represent time-based data (e.g., monthly attendance or sales).
3	15-01-26	16-01-26	Perform data representation vs data presentation for the same dataset using at least two different chart types.
4	15-01-26	16-01-26	Identify the purpose of data visualization for a given dataset and use a donut chart.
5	22-01-26	22-01-26	Explain the seven stages of data visualization using a simple example.
6	22-01-26	23-01-26	Create one basic visualization using any data visualization tool (Excel / Google Sheets / Tableau / Power BI).
7	29-02-26	30-02-26	Create an area chart for time-series data such as yearly profit or rainfall data.
8	05-02-26	06-02-26	Draw a scatter plot to show the relationship between two variables (e.g., study hours vs marks).
9	12-02-26	13-02-26	Create a pivot table and a pivot chart using a simple dataset.
10	12-02-26	13-02-26	Design a tree map to represent category-wise data (e.g., department-wise student count).
11	26-02-26	27-02-26	Draw a simple node-link diagram to show relationships (e.g., friends in a class).
12	26-02-26	27-02-26	Visualize the correlation between different numeric variables in a dataset using a color-coded matrix (heatmap).
13	05-03-26	06-03-26	Create a time-series visualization with multiple variables and analyze the observed patterns.
14	05-03-26	06-03-26	Perform text visualization by counting word frequency from a paragraph and plotting it.
15	12-03-26	13-03-26	Create a simple multivariate chart using more than two variables (e.g., marks in three subjects).
16	12-03-26	13-03-26	Study a small case example where visualization helps in understanding data easily.
17	19-03-26	20-03-26	Create a live-like data visualization using continuously updated values (manual update is allowed).
18	19-03-26	20-03-26	Write basic rules or best practices for creating good visualizations with examples.
19	02-04-26	27-03-26	Create a basic chart to show open and closed ports from given data.
20	02-04-26	27-03-26	Create a firewall log visualization to identify suspicious traffic patterns.
21	09-04-26	10-04-26	Design an intrusion detection log visualization and highlight possible attacks.

22	09-04-26	10-04-26	Write a program to implement sentiment analysis on the live tweet data using Python libraries.
23	16-04-26	17-04-26	Write a program to create interactive visualizations to monitor device activity for Pre-processed simulated IoT data streams in real-time.
24	16-04-26	17-04-26	Design a secured data visualization system architecture and explain the security measures used.